Bara M. Zalloum

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Education

University of Illinois Champaign, IL
Bachelor of Science in Computer Science Expected May 2027
Relevant coursework: Game Development, Computer Architecture, Data Structures & Algorithms, Database Systems

Experience

Central Illinois Mosque and Islamic Center (CIMIC) Urbana, IL
IT & AV Systems Support Aug 2025 – Present

- Built trigger-based automation systems (Home Assistant, Wake-on-LAN, MacroDroid) that orchestrate displays, channels, and livestream equipment, eliminating manual AV setup for weekly services
- Administer servers, IP cameras, and streaming infrastructure, diagnosing and resolving technical issues live during events

Rosecrance Behavioral Health Champaign, IL
Engagement Specialist May 2025 – Present

- Recorded 2,500+ service contacts with 1,750 unique clients across a three-county service area, coordinating care linkage between inpatient, outpatient, and community providers in a HIPAA-regulated environment
- Designed and maintained a data tracking system logging crisis events, response types, and outcomes, improving team visibility and operational efficiency

Confidential Technology Company IL
Software Engineer Intern May 2023 – Aug 2023

- Designed and implemented a centralized framework for team ecosystem documentation using Markdown, HTML, and JavaScript, improving accessibility and organization across the engineering team
- Developed configurable widgets enabling dynamic content creation and customization without predefined data sources

Projects

Maze / Puzzle Game – Team Final Project Champaign, IL
Unreal Engine, C++ • CS 415 Game Development 2026

- Designed and built a maze/puzzle game in Unreal Engine (C++) as the final project of a 4-person team, implementing core gameplay mechanics and puzzle-driven level layouts
- Collaborated across design and programming roles, integrating assets into playable scenes and iterating on gameplay through playtesting and feedback

Single-Player Game with AI Opponents Champaign, IL
Unreal Engine, C++ • CS 415 Game Development 2026

- Built a single-player game in Unreal Engine (C++) with AI-driven opponents that react to the player, implementing gameplay systems, game feel, and difficulty balance
- Debugged and optimized gameplay code for stability and performance

Endless Runner Champaign, IL
Unreal Engine, C++ • CS 415 Game Development 2026

- Developed an endless runner in Unreal Engine (C++) with procedurally spawned obstacles, an increasing difficulty curve, and collision and scoring systems

Selected Software Projects Champaign, IL
React, TypeScript, Vue 3, Python 2026

- Al-Baqaa: a production community obituary platform (React 18, TypeScript, OpenAPI 3.0, Node.js) with 6-language internationalization and GitHub Actions CI/CD
- Campus Housing: a Vue 3 apartment aggregator whose Python ETL pipeline scrapes 4 landlord portals and geocodes 174 listings, refreshed weekly by GitHub Actions

Technical Skills

Languages: C++, TypeScript, JavaScript, Python, Java, SQL, HTML/CSS, Bash
Game Development: Unreal Engine, Unreal Editor, Gameplay Mechanics, Enemy AI, Procedural Generation, Level Design, Playtesting & Iteration, Debugging & Optimization
Frameworks & Tools: React, Vue 3, Node.js, Git, GitHub, GitHub Actions (CI/CD), Linux/UNIX Shell, Agile/Scrum, AI-Assisted (Agentic) Development